



GOVERNMENT POLYTECHNIC, AMRAVATI
(AN AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)
CURRICULUM DEVELOPMENT CELL

PROGRAMME TITLE: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE: CM5474

COURSE TITLE: LINUX OPERATING SYSTEM

TEACHING SCHEME:

LEVEL OF COURSE	PRERE-QUISITE	WEEKLY CONTACT HRS.			TOTAL CREDITS	TOTAL WEEKS	TOTAL CONTACT HOURS		
		L	T	P			L	T	P
V	FC5462	03	-	02	05	16	48	-	32

EXAMINATION SCHEME:

THEORY(Marks)					PRACTICAL(Marks)		TOTAL (Marks)
ESE PAPER HRS.	ESE		PA	TOTAL	ESE	PA	
3Hrs	MAX.	70	30*	100	25#	25^	150
	MIN.	28	---	40	10	10	

@: Internal Assessment #: External assessment-Practical based \$: online examination

(*) Under the Theory PA, Out Of 30 Marks, 20Marks is the Average of Two Tests and 10 Marks are for Micro project-

(^)^ under practical PA Continuous Assessment of Practical Work is to be done by Course Teacher as per CDC norms.

For the courses having only practical examination, PA has two parts (i) Continuous Assessment of Practical work - 60% and (ii) microproject-40%.

1. RATIONALE:

Operating System is the interface between the user and the computer system. Nowadays LINUX is one of the most widely used operating system. Knowledge of LINUX operating system is essential as it provides many features such as multitasking, multiuser, security etc. which are mainly used in both server and workstation systems. So, this course will enable the students to inculcate the basics of LINUX Operating System, writing Shell scripts as well as administer the network.

2. COURSE OUTCOMES (COs)

At the end of this course, student will be able to: -

1. Install Linux Operating System.
2. Evaluate the basic linux OS commands and its utilities.
3. Develop Shell program for solving different problems.
4. Maintain Linux Operating System.
5. Manage Disk related operation in Linux operating System.
6. Maintain Linux Network Services.

3. DETAILED CONTENTS: THEORY

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topic and Sub-topics	CO No.	Hrs	Marks
Unit 1. Features of Linux Operating System	1a Describe the use of given system call of Linux OS. 1b Determine the file type on the basis of first character of Is command output for the given computer system. 1c Outline salient features of the given Linux shell.	1.1 Features of Open Source Operating Systems, Core Linux Distributions, Architecture, OS Services, System Calls, Run Levels. 1.2 File System: Hierarchical File System, File System features. 1.3 Shell: Login into the system, Concept of Shell, Various Linux Shell and their Features.	1	08	10
Unit 2. Linux Commands and Utilities	2a Classify the given command as internal or external. 2b Use the relevant command to produce the specified output. 2c Use relevant file and directory command(s) to perform the specified operation. 2d Apply the specified permissions to file and directory	2.1 Locating Commands, Internal & External Commands, Arguments, Options & Filenames, Online help 2.2 General Purpose Utilities cal, date, who, whoami, tty, uname, passwd, echo, tput, be,script, wall, write, mail. 2.3 Navigating the File System Concepts: Files, Directories, Paths, Home Directory, Parent-Child Relationship, Handling Command.-pwd, cd, mkdir, rmdir,ls Ordinary Files --_, handling commands: cat, cp,rm,mv, - cmp,comm,diff 2.4 File Attributes : File Permissions listing file, permissions, chmod Command 2.5 grep Family: Regular expressions, grep, egrep, fgrep, tr.	2	11	14
Unit 3. Shell Program	3a Use vi editor in the specified mode to carry out the given operation.	3.1vi Editor: Modes of vi, commands in various modes - creating,	3	11	16

m ing	3b	Apply relevant wild card for the given pattern matching.	editing, saving and quitting 3.2 Shell, sh Command, Pattern Matching-the Wild Cards, Escaping-the Backslash(\), Quoting, Redirection, Pipes, Tees, Command Substitution, Shell Variables 3.3 Shell Programming: Shell Scripts, read Statement, Command Line Arguments Positional Parameters, Exit Status of Command, Logical Operators && and , exit Statement, if and case Statements, expr Statement, while, until and for Statements, Sample Validation & Data Entry. 3.4 Simple Scripts, Scripts Using Simple Commands			
	3c	Create conditional statement using logical and relational operators to implement the given criteria.				
	3d	Write shell scripts for the given problem.				
Unit 4. Basic Linux System Administ ration	4a	Write procedure to perform the given task of System Administrator.	4.1 System Administration: Role of Administrator, root-Administrator's Login, su: Acquiring superuser Status, Administrator's Privileges- passwd Commands, Task Scheduling using cron, Maintaining Security. 4.2 Operations: Startup and shutdown, System run levels 4.3. User management: User configuration and password file, Managing Users and Groups.	4	06	10
	4b	Explain purpose of using the given run level.				
	4c	Write procedure to perform the given operation for managing the users/groups.				

Unit	Unit Outcomes	Topics and subtopics	Co No	Hrs	Marks
Unit 5. Basic Linux Disk management	5a Use relevant command as per the given disk management operation. 5b Managing Linux disks and the file systems that reside on them 5c Creation and deletion of disk partition.	5.1 Managing Disk Space: df, du, find command-Locating files dd, Command-Copying Disks, disk management-RAID. 5.2 Backups: Need of backup, cpio & tar commands. 5.3 Linux commands for disk management, fdisk, pydf, parted, lsblk, sfdisk, cfdisk. 5.4 Creation and deletion of partition in linux, formatting partition :mkfs	5	06	10
Unit 6. Basic Network Management	6a Configure the given TCP/IP settings in a network. 6b Configure the given setting in DHCP Server/Client. 6c Write method to configure setting in firewall to apply the given network security feature. 6d Use specified utility/software for Network Intrusion	6.1 IP address configuration: TCP/IP Network address, TCP/IP Configuration files, Network Interfaces and Routes :if config, route, ping, netstat, tcpdump commands 6.2 DHCP Server Configuration Configuring DHCP Client and Server, Dynamic Address, Fixed Addresses. 6.3 NIS, NFS, SAMBA introduction. 6.4 Firewall and Internet Security: Limiting Network Services, Designing Firewall. 6.5. Network Intrusion Detection: Host based Intrusion Detection Software using tripwire any relevant utility	6	06	10

4. LIST OF PRACTICALS:

Sr No.	PRACTICAL OUTCOMES (PrOs)	CO NO.
1	Install and configure Linux operating system environment.	1
2	Use pipe to concatenate the General Purpose Linux command.	2
3	Use pattern Searching using grep family commands.	2
4	Write a Shell script using following Control Structures : a) if then else structure and nested if then Structure.	3

	b) Case Statement	
5	Write a Shell script using expr to perform arithmetic Operations.	3
6	Write a Shell script using test command to check a) Two variables using -eq, -ge, -gt, -le, -it, -lle. b) Existence of file, file as a directory, file size greater than zero.	3
7	Write a login Shell script to perform a) Verify that your Shell, Looking at Current Values b) Editing Current Values, Test New Prompts.	3
8	Manage users and groups in Linux as a Super user,.	4
9	Manage various administrative privilege commands and startup and shut down of Linux OS.	4
10	Use relevant command as per the given disk management operation.	5
11	Create and delete of disk partition.	5
12	Configure TCP/IP settings and perform ICMP commands such as traceroute, ping.	6

Note

- The entire above listed practical's need to be performed compulsorily, so that the students reach the 'Precision level of Dave's Psychomotor Domain Taxonomy.
- The Process and Product related skills associated with each practical outcome shall be assessed on basis of following performance indicators.

Sr. No.	Performance Indicators	Weightage in %
a	Configuration of Linux operating system	20
b	Correctness of Executing various commands	20
c	Writing and executing shell script to get desired output	20
d	Debugging the program	15
e	Submit journal report in time	15
f	Question on term work	10
	Total	100

The above PrOs also comprise of the following social skills/attitudes which are Affective Domain Outcomes (ADOs) that are best developed through the laboratory/field based experiences:

- Follow safety practices.
- Practice good housekeeping.
- Demonstrate working as a leader/a team member.
- Maintain tools and equipment.
- Follow ethical Practices.

The ADOs are not specific to any one PrO, but are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he undertakes a series of practical experiences over a period of time. Moreover, the level of achievement of the ADOs according to Krathwohl's 'Affective Domain Taxonomy' should gradually increase as planned below:

- 'Valuing Level' in 1st year
- 'Organizing Level' in 2nd year
- 'Characterizing Level' in 3rd year

5. SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student- related co-curricular activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5

pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a. Prepare journal of practicals.
- b. Undertake micro-projects.

6. SUGGESTED INSTRUCTIONAL STRATEGIES

Following are suggested instructional strategies, which the teacher can adopt for the attainment of the various outcomes in this course:

- a. Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b. The teacher needs to ensure to create opportunities and provisions for **co-curricular activities**.
- c. Guide student(s) in undertaking micro-projects.
- d. Demonstrate students thoroughly before they start doing the practice
- e. Use different instructional strategies in classroom teaching.
- f. Observe continuously and monitor the performance of student in lab.
- g. About 10-15% of the topics/sub-topics/contents which is relatively simpler or descriptive in nature may be given to the students for self-directed learning and assess the development of the COs through classroom presentations. Keep the record of the topics/sub-topics/contents given to the students.

7. SUGGESTED MICRO-PROJECTS.

Only one micro-project is to be undertaken by a student in the beginning of the semester. S/he ought to submit it by the end of semester. Each micro-project should include two or more Cos. The micro-project could be industry application based, internet based, laboratory based, field based and survey based. In the all semesters, it could be group-based (group 5-6 student) to build up skill and confidence in every student. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The student ought to submit micro-project by the end of the semester to develop the industry oriented COs.

A suggestive list of micro-projects is given here. Similar micro-projects could be added by the concerned faculty:

- a) Configure following Common Services (Client AND Server)
- b) DHCP, DNS, LDAP, Email (SMTP, POP, IMAP)
- c) Build a NAS
- d) Configure NFS, Samba
- e) Configure Proxy Server, Web server, Squid, Apache
- f) Outline salient features of the given Linux shell.
- g) Write a grep/egrep script to find the number of words character, words and lines in a file.
- h) Installation of Virtual Box (VMware) on a PC having other operating system.
- i) Shell script program to count number of files in a Directory.
- j) Shell script program to copy contents of one file to another.
- k) Create directory, write contents on that and Copy to a suitable location in your home directory.

8. MAJOR EQUIPMENTS/INSTRUMENTS REQUIRED

Sr No.	Equipment Name with Broad Specification	Practical No.
1	Computer system (Any computer system with basic configuration)	All
2	Linux or alike operating system such as Ubuntu, CentOS or any other.	

9. SUGGESTED SPECIFICATION TABLE FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Marks per Unit	Distribution of Theory Marks		
			R Level	U Level	A Level
1	Features of Linux Operating System	10	02	04	04
2	Linux Commands and Utilities	14	02	04	08
3	Shell Programming	16	04	04	08
4	Basic Linux System Administration	10	02	04	04
5	Basic Linux Disk management	10	02	04	04
6	Basic Network Management	10	02	04	04
Total		70	14	24	32

R= Remember, U= Understanding, A=Application and above (*Bloom's Revised taxonomy*)

Note: This specification table provides general guidelines to assist student for their learning and to teachers to teach and assess students with respect to attainment of UOs. The actual distribution of marks at different taxonomy levels (of R, U and A) in the question paper may vary from above table.

10. SUGGESTED LEARNING RESOURCES:

Sr. No.	Title Of Book	Author	Publication
1.	Unix Concept and Programming	Das, Sumitabha	McGraw Hill education, New Delhi, 2015, ISBN: 978-0070635463
2.	Linux Command Line and Shell Scripting Christine Bible, 3rd Edition	Blum, Richard, Bresnahan, Christine	Wiley Publication, New Delhi, 2015, ISBN:- 978-1-118-98384-3
3.	Red Hat: The Complete Reference Enterprise Linux and Fedora Edition	Piterson, Richard	McGraw Hill education, New Delhi, ISBN:-0-7-058709-4
4.	Red Hat Linux Networking and System Administration	Coiling, Terri & Wall, Kurt	Red Hat ISBN: 0-7645-3632-X ISBN: 978-007063546

11. SUGGESTED SOFTWARE/LEARNING WEBSITES

- <https://www.linux.com/docs/tools-references/linux-system-administration-basics>.
- <http://training.linuxfoundation.org/free-linux-training>.
- <https://www.javatpoint.com/linux-tutorial>
- <http://www.tutorialspoint.com/linux-tutorials/linux/1>
- <https://www.digitalocean.com/community/tutorials/how-to-customize-your-bash-prompt-on-a-linux-vps>
- https://spoken-tutorial.org/tutorial-search/?search_foss=Linux&search_language=English

12. COURSE CURRICULUM DEVELOPMENT COMMITTEE:

SR. NO.	NAME	DESIGNATION	INDUSTRY/INSTITUTE
1	Dr.P.R.Satav	Lecturer in Computer Engineering.	Government Polytechnic Amravati
2	Smt.V.R.Rathod	Lecturer in Computer Engineering.	Government Polytechnic Amravati

Government Polytechnic, Amravati Programme Board of Studies Computer Engineering has approved the above course curriculum on 30/12/2020 and is adopted for Computer Engineering Programme.

CHAIRMAN
PROGRAMME BOARD OF STUDIES,
COMPUTER ENGINEERING
GOVERNMENT POLYTECHNIC, AMRAVATI.

The General Board of Studies has approved the above course curriculum on 06/02/21
The Governing Body has approved the above course curriculum on 13/08/2021